

Logistic Regression Models with R

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1 Loading Data

We will use the *Palmer Penguin* data set available directly from the `palmerpenguins` package. You can learn more about the data at allisonhorst.github.io/palmerpenguins.

Today, we will explore whether **biometrics can be used to predict sex**.

```
# install.packages("palmerpenguins")
library(palmerpenguins)
library(ggplot2)
library(car)

dat <- as.data.frame(penguins) # rename dataset
str(dat) # structure of dataset
```

Let's focus on one species.

```
dat_gen = subset(dat, species == "Gentoo" & !is.na(sex))
```

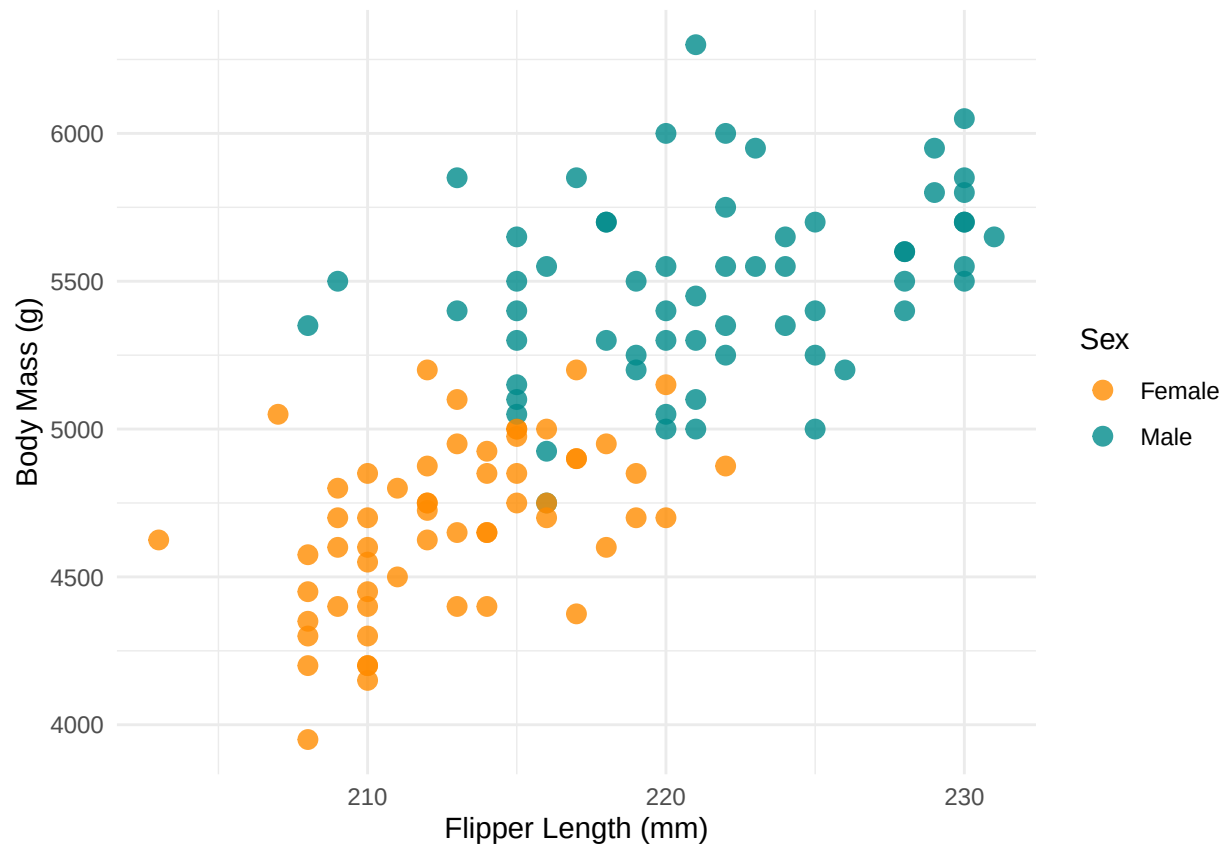
2 Exploring the data

Which one of the variables included in the data may contribute be associated with sex?

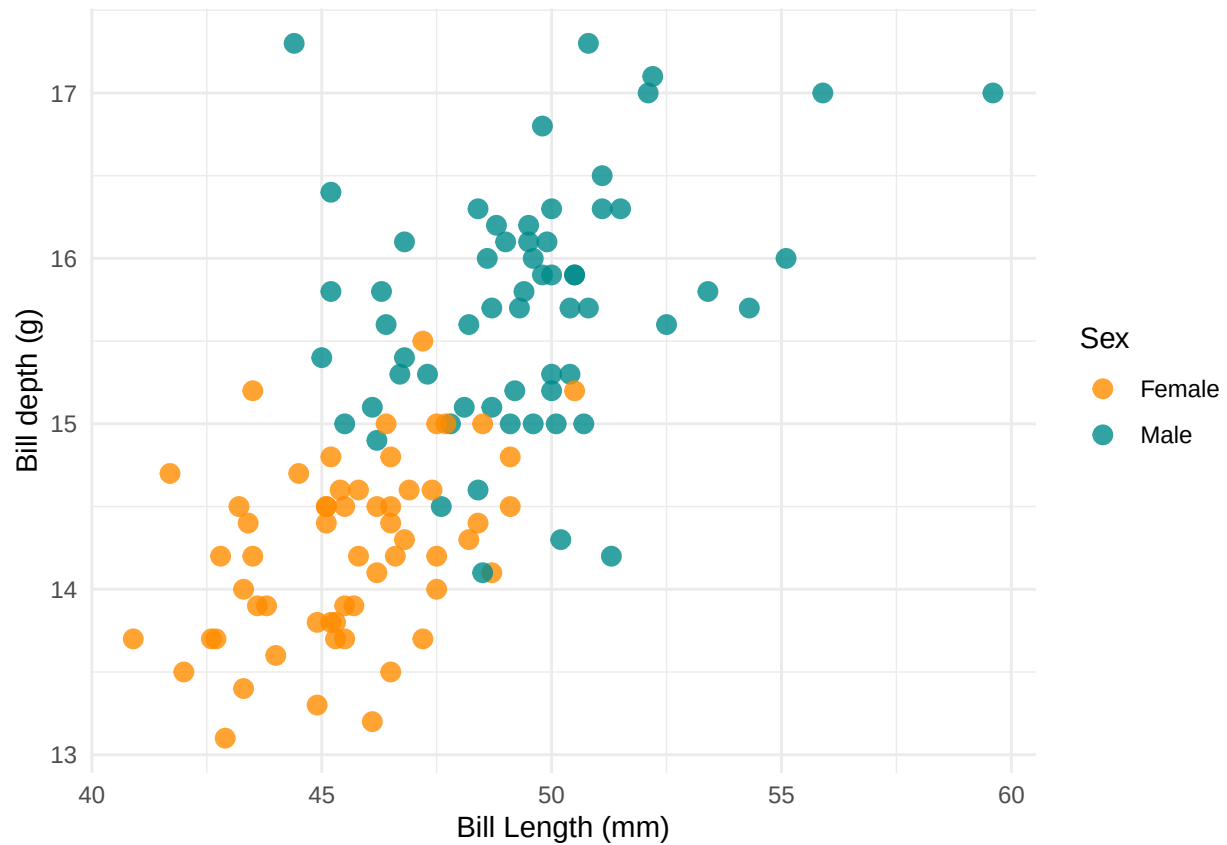
2.1 Plot

```
ggplot(dat_gen, aes(x = flipper_length_mm, y = body_mass_g, color = sex)) +
  geom_point(size = 3, alpha = 0.8) +
  scale_color_manual(values = c("darkorange", "cyan4"),
                    name = "Sex",
                    labels = c("Female", "Male", "Unknown")) +
  scale_shape_manual(values = c(16, 17, 15),
                    name = "Species") +
```

```
labs(
  x = "Flipper Length (mm)",
  y = "Body Mass (g)"
) +
theme_minimal()
```



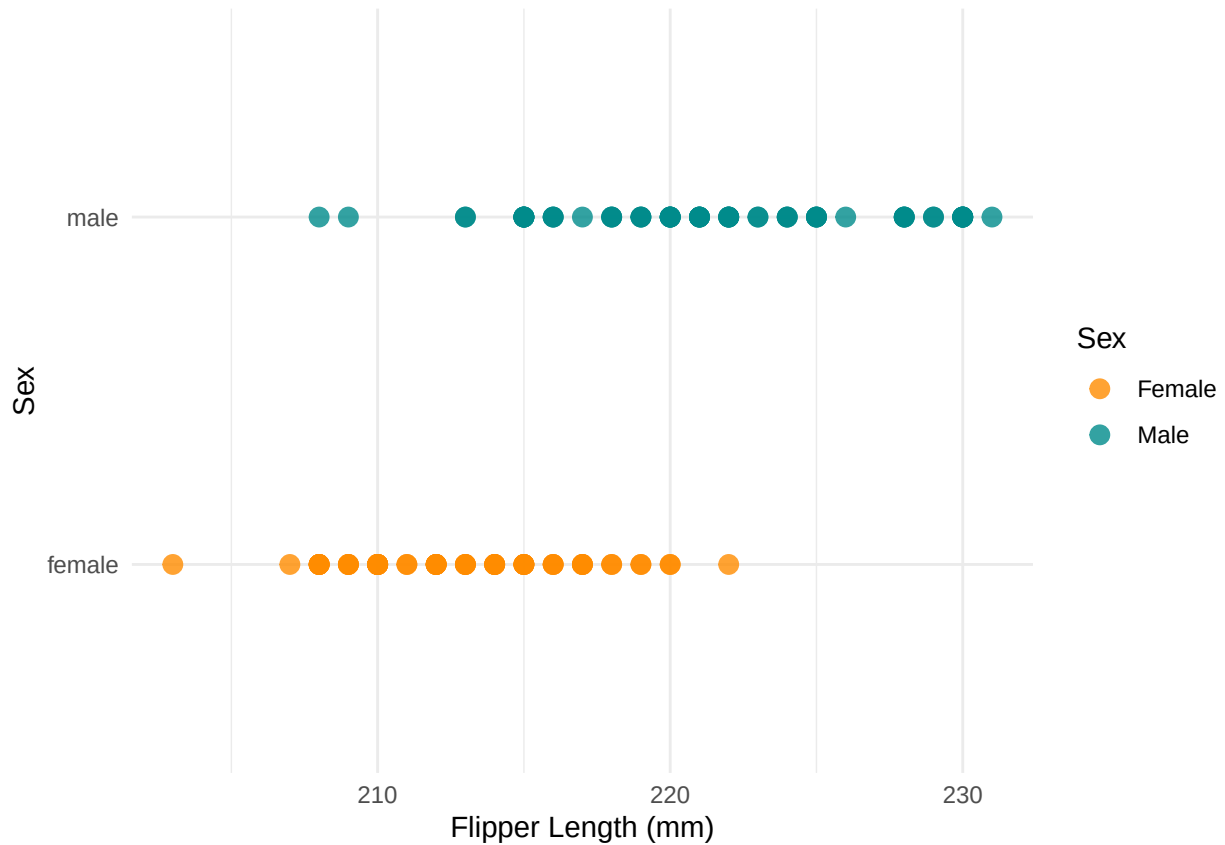
```
ggplot(dat_gen, aes(x = bill_length_mm, y = bill_depth_mm, color = sex)) +
  geom_point(size = 3, alpha = 0.8) +
  scale_color_manual(values = c("darkorange", "cyan4"),
    name = "Sex",
    labels = c("Female", "Male", "Unknown")) +
  scale_shape_manual(values = c(16, 17, 15),
    name = "Species") +
  labs(
    x = "Bill Length (mm)",
    y = "Bill depth (g)"
  ) +
  theme_minimal()
```



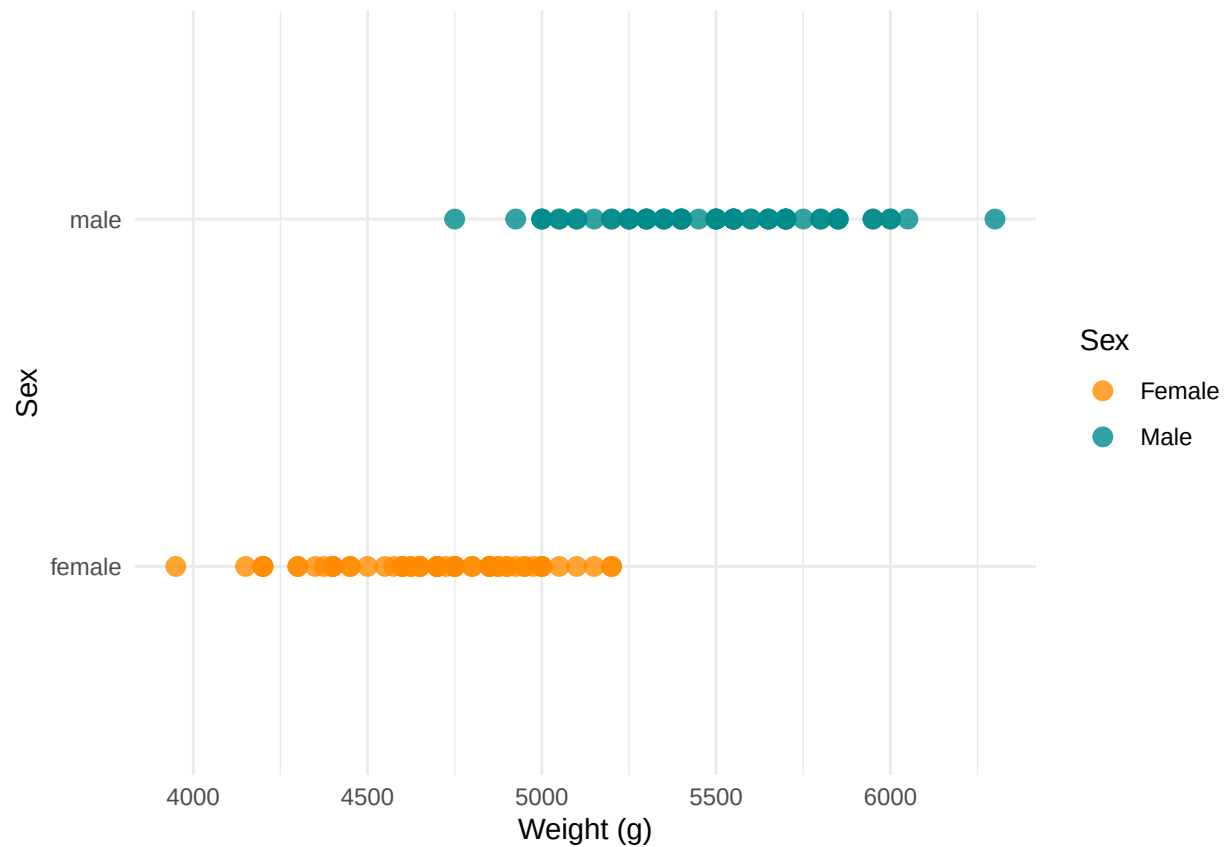
3 Logistic Regression to Identify Predictors of Sex

3.1 Re-plot

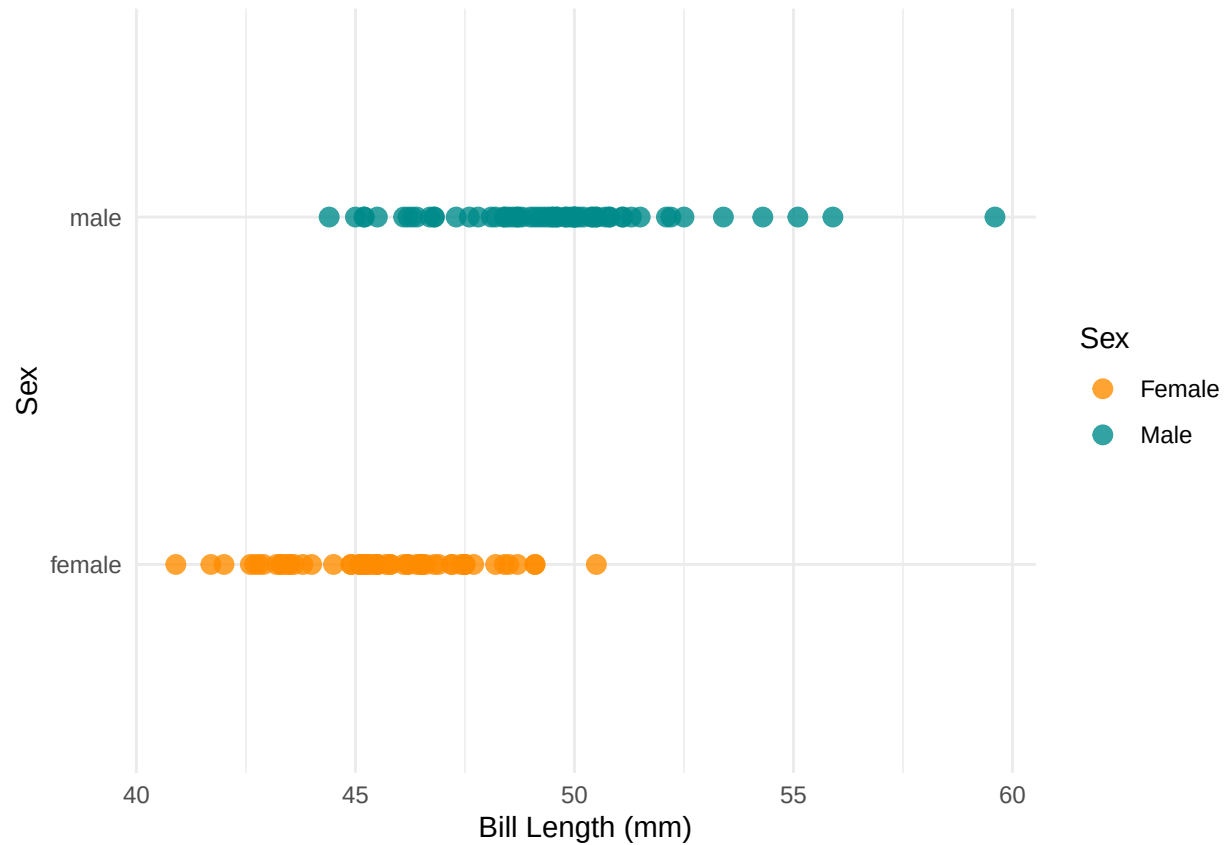
```
ggplot(dat_gen, aes(x = flipper_length_mm, y = sex, color = sex)) +
  geom_point(size = 3, alpha = 0.8) +
  scale_color_manual(values = c("darkorange", "cyan4"),
                    name = "Sex",
                    labels = c("Female", "Male", "Unknown")) +
  scale_shape_manual(values = c(16, 17, 15),
                    name = "Species") +
  labs(
    x = "Flipper Length (mm)",
    y = "Sex"
  ) +
  theme_minimal()
```



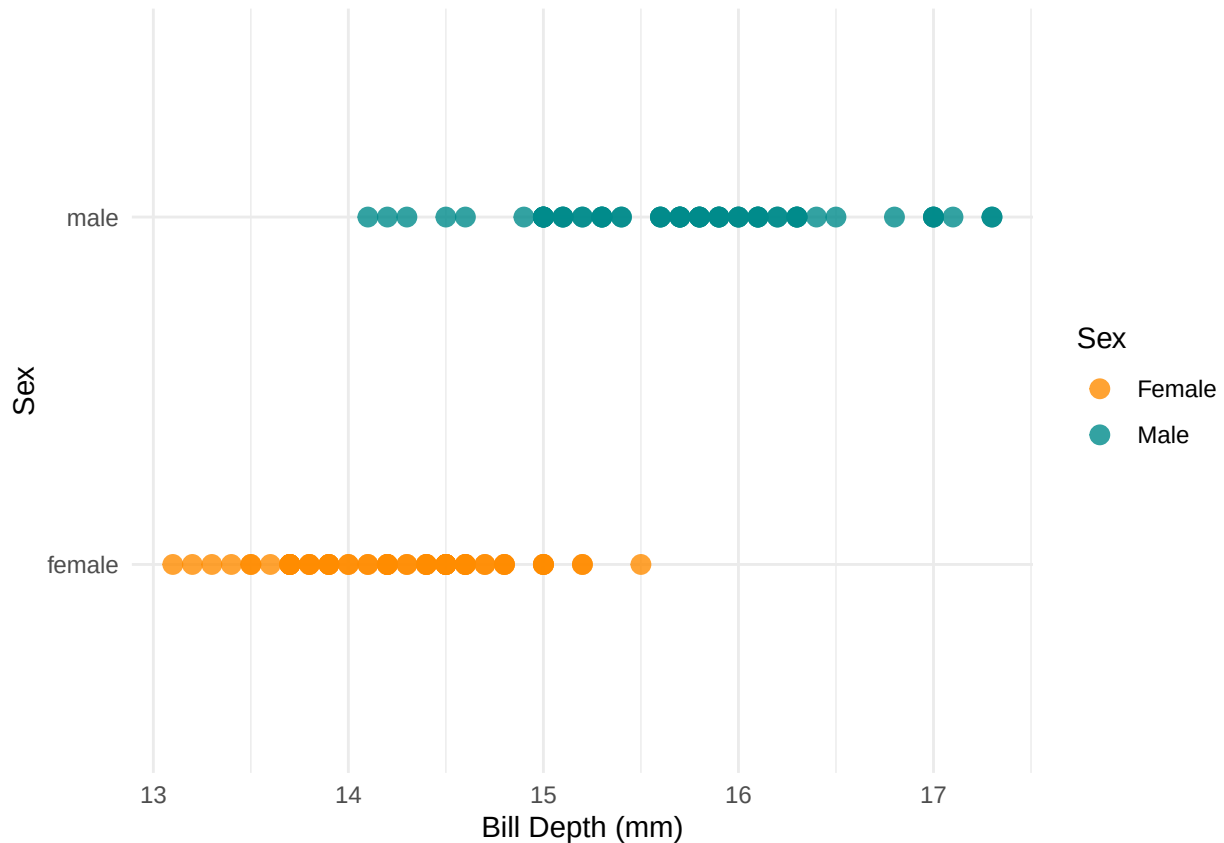
```
ggplot(dat_gen, aes(x = body_mass_g, y = sex, color = sex)) +
  geom_point(size = 3, alpha = 0.8) +
  scale_color_manual(values = c("darkorange", "cyan4"),
                     name = "Sex",
                     labels = c("Female", "Male", "Unknown")) +
  scale_shape_manual(values = c(16, 17, 15),
                     name = "Species") +
  labs(
    x = "Weight (g)",
    y = "Sex"
  ) +
  theme_minimal()
```



```
ggplot(dat_gen, aes(x = bill_length_mm, y = sex, color = sex)) +
  geom_point(size = 3, alpha = 0.8) +
  scale_color_manual(values = c("darkorange", "cyan4"),
                     name = "Sex",
                     labels = c("Female", "Male", "Unknown")) +
  scale_shape_manual(values = c(16, 17, 15),
                     name = "Species") +
  labs(
    x = "Bill Length (mm)",
    y = "Sex"
  ) +
  theme_minimal()
```



```
ggplot(dat_gen, aes(x = bill_depth_mm, y = sex, color = sex)) +
  geom_point(size = 3, alpha = 0.8) +
  scale_color_manual(values = c("darkorange", "cyan4"),
                    name = "Sex",
                    labels = c("Female", "Male", "Unknown")) +
  scale_shape_manual(values = c(16, 17, 15),
                    name = "Species") +
  labs(
    x = "Bill Depth (mm)",
    y = "Sex"
  ) +
  theme_minimal()
```



Does a linear relationship seem reasonable?

4 Statistical Test

4.1 Assumptions

4.1.1 Independance

Observations from each population are independent of one another. This is likely the case (would be determined by the study design).

4.1.2 Linearity

Linearity between the response variable odds and the explanatory variables. Visible from the scatter plots above.

4.1.3 Absence of Collinearity

Absence of collinearity between the explanatory variables. Suspicious from the scatter plots above.

4.2 Model

```
# Fit model
mod = glm(sex ~ flipper_length_mm + bill_length_mm + bill_depth_mm + body_mass_g,
          data = dat_gen,
          family = "binomial")
```

4.2.0.1 Model Fitting

4.2.0.2 Additional Assumption Checks Double checking for the absence of collinearity.

```
# Variance Inflation Factor  
vif(mod)
```

The variance inflation factor is low (< 2) for every variable so we can keep all of them.

```
mod = glm(sex ~ flipper_length_mm,  
          data = dat_gen,  
          family = "binomial")  
summary(mod)  
  
mod = glm(sex ~ flipper_length_mm + bill_length_mm,  
          data = dat_gen,  
          family = "binomial")  
summary(mod)  
  
mod = glm(sex ~ flipper_length_mm + bill_length_mm + bill_depth_mm,  
          data = dat_gen,  
          family = "binomial")  
summary(mod)  
  
mod = glm(sex ~ flipper_length_mm + bill_length_mm + bill_depth_mm + body_mass_g,  
          data = dat_gen,  
          family = "binomial")  
summary(mod)
```

4.2.0.3 Model Selection

```
mod = glm(sex ~ flipper_length_mm + bill_length_mm + bill_depth_mm,  
          data = dat_gen,  
          family = "binomial")  
summary(mod)
```

4.2.0.4 Final Model

```
# Calculate R-squared  
deviance <- summary(mod)$deviance  
null_deviance <- summary(mod)$null.deviance  
rsquared <- 1 - (deviance / null_deviance)  
rsquared
```

4.2.0.5 Coefficient of Determination Based on a logistic regression with forward selection, we found that, if p is the probability of being a male, we can write:

$$\text{logit}(p) = \log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 * \text{flipper} + \beta_2 * \text{bill.length} + \beta_3 * \text{bill.depth}$$

With $\beta_0(SE) = -111.94(25.40)$, $\beta_1(SE) = 0.19(0.10)$, $\beta_2(SE) = 0.57(0.22)$ and $\beta_3(SE) = 2.94(0.77)$

The R^2 correlation coefficient of 0.70, indicates a moderate predictive power (a lot of the variability remains unexplained).